

MODULE-03

Importance of transportation

The evolution and advancements in transportation facilities have been, closely linked with the development of human being throughout the history of the world. Transportation contributes to the economic, industrial, social and cultural development of any country. Transportation is vital for the economic development of any region since every commodity produced, whether it is agricultural or industrial product needs to be transported at various stages from production to distribution. At the production stage, transportation is required for carrying raw materials like seeds, manure, coal, steel, machines, component parts. At the distribution stage, transportation is required from the production centres like the farms and factories to the marketing centres and later to the retailers and to the consumers. Inadequate transportation facilities retard the process of socio-economic and cultural development- of the country. Development of adequate transportation system in a country indicates its economic growth and progress in social development.

In the present day concept, the main objective of a good transportation system is to provide safe, economical, efficient transportation facility for the travel of passengers and transportation of goods.

Modes of Transportation

The following are the major modes of transportation:

- (i) Roadways or highways for road transportation
- (ii) Railways for rail transportation
- (iii) Waterways for water transportation
- (iv) Airways for air transportation

Road Transportation

- Transportation by road system is the only mode which could give maximum flexibility of service from origin to destination, to one and all, Various class of vehicles such as car, bus, truck, two- wheeler, etc. may be permitted to make use of the road.
- Apart from the road vehicles, pedestrians also make use of the facilities of the road system. Road transport mode has the maximum flexibility for travel with reference to choice of the route, direction, time and speed of travel.
- Road transportation is the only mode which caters for the movement of passengers and goods independently right from the place of origin up to the destination of any trip along the land.

- In other words, it is possible to provide door to door service only by road transport. The other three modes, viz., railways, waterways and airways have to depend on transportation by road for the service to and from their respective terminals, viz., railway stations, harbours and airports.
- Therefore, road network is essential to serve as feeder system for all, other modes of transportation and also to supplement them. Thus it is necessary to develop a well-planned road network throughout the country so as to serve both as an independent transport system and also as feeder transport for the other three modes of transport.

Rail Transportation

- The concept of rail transportation is movement of multiple wagons or a train of wagons or passenger bogies fitted with steel wheels running over two parallel steel rails of the railway track.
- The resistance to traction along the railway track for the movement of the steel wheels of the rail wagons is much lower than that along the more uneven road surface for the movement of road vehicles with rubber tyres, under identical conditions of speed and atmospheric factors.
- The energy requirement to haul unit load through unit distance by the railway is only a fraction (one fourth to one sixth) of that required by road. Therefore, full advantage of rail transportation should be taken for the transportation of bulk goods and passengers, especially for long distance on land routes and where the railway tracks are available.
- Rail transportation between the railway stations could be economical and advantageous both for the passengers and also for transportation of goods on long haul distances. The rail ways can transport a large number of goods at a passengers or a large quantity of goods at a time.
- The railway also serves as mass rapid transport for the commuters to travel from the suburban areas to the urban centres and for travel within the urban area of large cities. Different types of locomotives with steam or diesel engines or electric locomotives along the electrified railway tracks have been used in India to haul the trains. However, steam locomotives utilising heat energy produced by burning coal has got almost extinct in our country due to poor energy efficiency and pollution.

Water Transportation

- Transportation by water offers minimum resistance to traction and therefore needs minimum energy to haul 'unit load through unit distance. The amount of energy

required to propel a vessel on water is much lesser than that required on land and is far lesser than that required to keep an aircraft air-borne.

- Therefore, water transportation is the most energy efficient. But, water transportation is very slow, which is the greatest disadvantage of this mode. In fact, this is the slowest among the four modes.
- Though some passenger traffic makes use of water transportation, the greatest user of this mode is bulk cargo of relatively low value, mainly because of the slow speed and low transportation cost for the bulk cargo.
- Waterway is needed to operate the ships and boats on sea routes between the harbours and ports. Harbours are terminals built with components such as docks, quays, wharfs, etc.
- The other facilities required are to provide shelter to the ships. And the crews and also facilities for loading, unloading and storage of cargo and for carrying out routine and major maintenance of the vessels. The branch of transportation engineering which deals with the planning, design, construction and maintenance of docks and harbours is called Harbour Engineering.
- Inland water transportation is made possible along the rivers, canals and lakes by boat and ferry service, which operate between small ports. Inland waterways, being of relatively shallow depth, the freight is mostly carried on barges and are hauled by towing boats.

Air Transportation

- Transportation by air is the fastest among the four modes. Air transport provides more comfortable and fast travel resulting in substantial saving in travel time for the passengers between the airports.
- The shipment of high-value' freight on long hauls is possible in the shortest time by air transport. Unlike other modes of transport, air transport allows continuous journey over the land and water, even across inaccessible places in between two airports.
- One of the limitations is that the energy required and the overall operating expenses for air transportation are the highest in comparison to the other modes. Another limitation is that operation of air transport is very much affected by weather conditions such as severe storms and thick fog.
- Airports are developed to provide facilities for the aircraft for take-off, landing, parking and carrying out maintenance works. Some of the essential components of an airport

are runway, taxiway, terminal facilities, visual aid, control and safety system. The branch of transportation engineering which deals with the planning, design, construction and maintenance of the airports to cater for the needs of the aircrafts that are expected to make use of these airports is called Airport Engineering.

ASSGINMENT- BASIC CHARACTERSTICS, ADVANTAGES AND DISADVANTAGES OF ALL MODES OF TRANSPORTATION

Classification of Roads

The roads are generally classified into two categories, depending on whether they can be used during different seasons of the year

(i) All-weather roads and

(ii) Fair-weather roads

- **All-weather roads** are those which are negotiable during all seasons of the year, except at major river crossings where some interruption to traffic is permissible up to a certain extent, but the road pavement should be negotiable during all weathers. On '**fair weather roads**' the traffic may be interrupted during monsoon season at causeways where streams may overflow across the road.
- Based on the type of the carriageway or the road pavement, the roads are classified as "**Paved roads**' and "**Un-paved roads**". The roads with a hard pavement surface on the carriageway are called 'paved roads. The roads without a hard pavement surface on the carriageway are called 'unpaved roads. Earth roads and gravel roads may be called unpaved roads.
- Based on the type of pavement surfacing provided, the roads may be classified as '**Surfaced roads**' and '**Un-surfaced roads**'. Road pavements with any type of bituminous surface or cement concrete are called surfaced roads. The roads which are not provided with a bituminous or cement concrete surfacing are called un-surfaced roads.

The roads are generally classified based on the following:

- (a) Traffic volume
- (b) Load transported or tonnage
- (c) Location and function

The classification based on traffic volume or tonnage have been arbitrarily fixed by different agencies and there may not be a common agreement regarding the limits for each of classification group, based on the traffic volume or flow, **the roads are classified as heavy, medium and low volume roads.** These terms are relative and so the limits under each class should be clearly defined and expressed as vehicles per day or annual average daily traffic', etc. Likewise, **the classification based on load or tonnage is also relative and the roads may be classified as class I, II etc. or class A, B. etc.** and the limits may be expressed in terms of tonnes per day.

Road classification based on location and function

The Nagpur Road Plan classified the roads in India based on location and function into following five categories and described below:

- (i) National Highways (NH)
 - (ii) State Highways (SH)
 - (iii) Major District Roads (MDR)
 - (iv) Other District Roads (ODR) and
 - (v) Village Roads (VR)
- **National Highways (NH)** are main highways running through the length and breadth of India, connecting major ports, foreign highways, capitals of large states and large industrial and tourist centres including roads required for strategic movements for the defence of India.
 - It was specified that national highways should be the frame on which the entire road communication should be based and that these highways may not necessarily be of same specification, but they must give an uninterrupted road communication throughout the country and should connect the entire road network. The responsibility of construction and maintenance of national highways was decided to be with the central government.
 - **State Highways (SH)** are arterial roads of a state, connecting the national-highways of adjacent state, district headquarters and important cities within the state and serve as the main arteries for traffic to and from district roads.

- These highways are considered as main arteries of commerce by road within a state or a similar geographical unit. In some places they may even carry heavier traffic than some of the national highways but this will not alter their designation or function. The NH and SH have the same design speed and geometric design specifications.
- **Major District Roads (MDR)** are important roads within a district serving areas of production and 'markets and connecting with other major roads or main highways of a district. The MDR has lower speed and geometric design specifications than NH/SH.
- **Other District Roads (ODR)** are roads serving rural areas of production and providing them with outlet to market centres, taluk headquarters, block development headquarters or other than roads. These are of lower design specifications than MDR.
- **Village Roads (VR)** are roads other to connecting villages or groups of villages with each the nearest road of higher category. It was specified that these village roads should be in essence farm tracks, but it was desired that the prevalent practice of leaving such tracks to develop and maintain by themselves should be replaced by a plan for a designed and regulated system.

Classification of urban roads

The road systems within urban areas are classified as Urban Roads and will form a separate category of roads to be taken care by the respective urban authorities.

The urban roads are classified as:

(a) Arterial roads

(b) Sub-arterial roads

(c) Collector streets and

(d) Local streets

Arterials and sub-arterials are urban roads primarily for through traffic on a continuous route, but the sub-arterials have a lower level of traffic mobility than the arterials. Collector streets provide access to arterial roads and they collect and distribute traffic from and to local streets which provide access to abutting property.

Basic components of Road

Cross Slope or Camber

- Cross slope or camber is the slope provided to the road surface in the transverse direction to drain off the rain water from the road surface. Drainage and quick disposal of water from the pavement surface by providing cross slope is considered important because of the following reasons:

- To prevent the entry of: surface water into the pavement layers and the subgrade soil through pavement; the stability, surface condition and the life of the pavement get adversely affected if the water enters in to the subgrade and the soil gets soaked.
- To prevent. the entry of water into the bituminous pavement layers, as Continued contact with water causes stripping of bitumen from the aggregates and results in deterioration of the pavement layer.
- To remove the rain water from the pavement surface as quickly as possible and to allow the pavement to get dry soon after the-rain.

The rate of camber or cross slope is. usually designate by 1 in n which means that the transverse slope is in ratio 1 vertical to n horizontal. Camber may also be expressed as a percentage. If the camber is x %, the cross slope is x in 100.

Width of Pavement or Carriageway

- The width of pavement or carriageway depends on (i) width of traffic lane and (ii) Number of lanes. The portion of carriageway width that is intended for one line of traffic movement is called a traffic lane.
- Therefore, considering all these factors, a width of 3.5 m per lane is generally provided on rural highways in India, if the pavement has two or more lanes. The lane width is increased to 3.75 m if there is raised kerb, such as foot path of urban roads. On single lane roads the width of carriageway is 3.75 whereas in low volume roads or village roads, the width may be lesser.

What is Carriageway in Road | IRC Recommendation



Medians/Traffic Separators

- In highways with divided carriageway, a median is provided between two sets of traffic lanes intended to divide the traffic moving in opposite directions.
- The Main function of the median is to prevent head-on collision between vehicles moving in opposite directions on adjacent lanes. The median is also called or traffic separator. The traffic separators used may be in the form of pavement markings, physical dividers or area separators.
- The medians/traffic separators may also serve the following functions:
 - i)To channelize traffic into streams at intersections
 - ii)To shadow the crossing and turning traffic
 - iii)To segregate slow traffic
 - iv)To protect pedestrians
- It is desirable to provide wide median/'traffic separators of 8 to 14 m width so that it is possible carry out future widening of the carriageway towards the median.
- The IRC recommends a minimum desirable width of 5.0 m for medians of rural highways, which may be reduced to 3.0 m where land is restricted.' On long bridges the width of the median may be reduced up to 1.2 to 1.5 m. The medians should normally be of uniform width on a particular road, but where change in width is unavoidable, a transition of 1 in 15 to 1 in 20 must be provided.



Kerbs

- Kerb indicates the boundary between the pavement and median or foot path or island or shoulder. It is desirable to provide kerbs on urban roads.



Road Margins

- The various elements included in the road margins are shoulder, guardrail, foot path, drive way, cycle track, parking lane, bus bay, lay-bye, frontage road and embankment slope.



Shoulders

- Shoulders are provided on both sides of the pavement all along the road in the case of undivided carriageway. The minimum shoulder width recommended by the IRC is 2.5 m.
The important functions of shoulders are:
- Shoulders provide structural stability and support to the edges of the flexible pavements, particularly to the granular pavement layers.
- The capacity of the carriageway and the operating speeds of vehicles increase if the shoulders are laid and maintained in good condition.
- Shoulders serve as emergency lanes for vehicle compelled to be taken out of the main carriageway or roadway.
- Shoulders also act as service lanes for vehicles that are disabled.



Guard rails

- Guard rails are provided at the edge of the shoulder when the road is constructed on a fill so that vehicles are prevented from running off the embankment, especially when the height of the fill exceeds 3 m.



Footpath or side-walk

- In order to provide safe facility to pedestrians to walk along the roadway, foot paths or side-walks are provided in urban areas where the pedestrian traffic is noteworthy and the vehicular traffic is also heavy.
- The absolute minimum width of foot path is 1.5 m and the desirable minimum width is 2.0 m; the width may be increased based on the pedestrian traffic volume.



Cycle tracks

- Cycle tracks are provided in urban areas where the volume of cycle traffic on the road is very high. A minimum width of 2 m is provided for the cycle track and the width may be increased by 1.0 m for each additional cycle lane.

Width of Formation or Roadway

- Width of formation or roadway is the sum of widths of pavement or carriageway including separators, if any and the shoulders.



Right of Way (ROW)–

- Whole length of the road is the right of way. It is the combination of carriage way, shoulder, margin, and other components of the road.



FLEXIBLE PAVEMENTS

1. Grain to grain load transfer
2. Initial cost is low
3. Joints are not required
4. Durability is less
5. Good subgrade is required
6. Temperature variation has no any effect on the stress variation
7. Life span is short ~ 15 years
8. Repair work is easy
9. Maintenance cost is high
10. Requires less curing time
11. Poor night visibility due to use of bitumen
12. No glare due to sunlight
13. Easy to locate the underground works like pipe location, etc.
14. Thickness is more
15. Design depends upon the subgrade strength
16. Stability depends upon the aggregate interlocking, particle friction and cohesion.
17. IRC 37



RIGID PAVEMENTS

1. Slab action takes place
2. Initial cost is high
3. Joints are required
4. Durability is high
5. Good subgrade is not required
6. Temperature variation effects the stress variation
7. Long life span ~ 30 years
8. Repair work is tough
9. Maintenance cost is low
10. Requires much curing time
11. Good night visibility
12. High glare due to sunlight
13. Difficult to do the underground works
14. Thickness is less
15. Design not depends on subgrade
16. Stability depends upon the joints between the slabs of concrete.
17. IRC 58



